

Introduction to Federated Learning

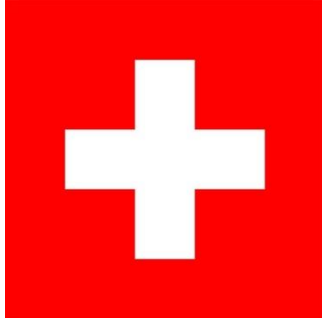
SIB course on Federated Learning in Bioinformatics

Daniele Malpetti, Lecturer-Researcher
*Dalle Molle Institute for Artificial Intelligence (IDSIA), SUPSI
Swiss Institute of Bioinformatics (SIB)*

Federated Learning

How everything started

Disclaimer



Federated Learning is **not**
something related to the
Swiss Confederation.

A basic definition

What is federated learning? Please give me a concise definition.



Federated learning is a collaborative machine learning method where a model is trained across multiple decentralized devices or servers that each hold local data, without sharing the data between them.



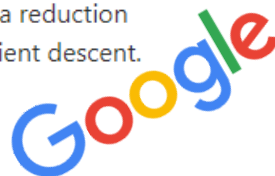
The origin of the name

Communication-Efficient Learning of Deep Networks from Decentralized Data

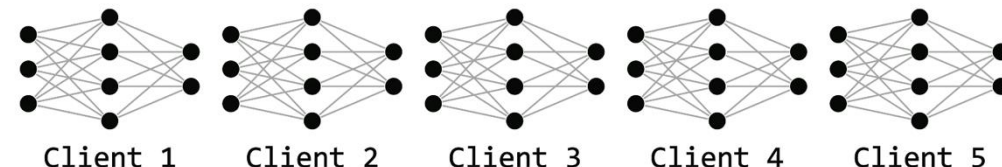
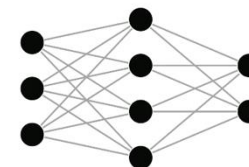
Brendan McMahan, Eider Moore, Daniel Ramage, Seth Hampson, Blaise Agüera y Arcas *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics*, PMLR 54:1273-1282, 2017.

Abstract

Modern mobile devices have access to a wealth of data suitable for learning models, which in turn can greatly improve the user experience on the device. For example, language models can improve speech recognition and text entry, and image models can automatically select good photos. However, this rich data is often privacy sensitive, large in quantity, or both, which may preclude logging to the data center and training there using conventional approaches. We advocate an alternative that leaves the training data distributed on the mobile devices, and learns a shared model by aggregating locally-computed updates. **We term this decentralized approach Federated Learning.** We present a practical method for the federated learning of deep networks based on iterative model averaging, and conduct an extensive empirical evaluation, considering five different model architectures and four datasets. These experiments demonstrate the approach is robust to the unbalanced and non-IID data distributions that are a defining characteristic of this setting. Communication costs are the principal constraint, and we show a reduction in required communication rounds by 10-100x as compared to synchronized stochastic gradient descent.



Server



FedAvg: averaging of weights

The origin of the name

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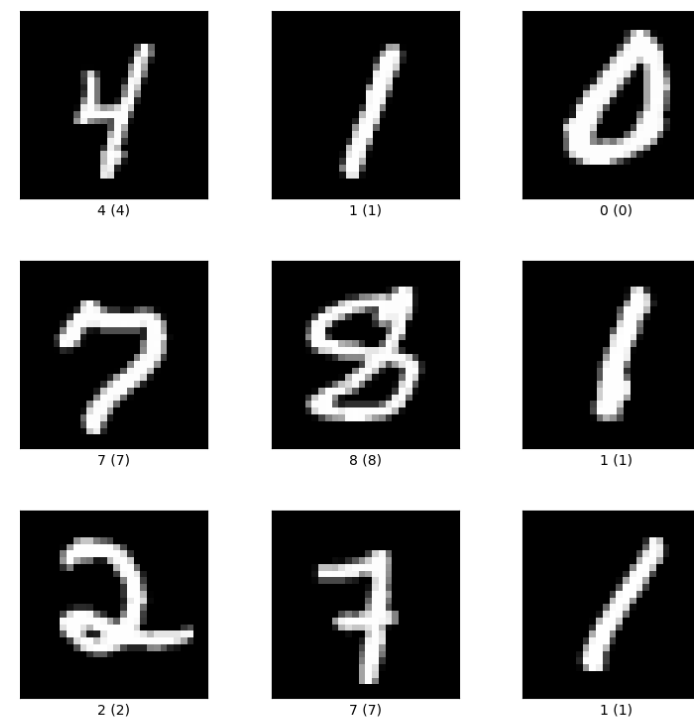
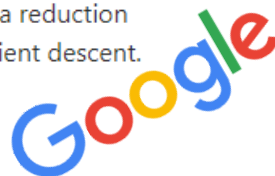


Image classification task

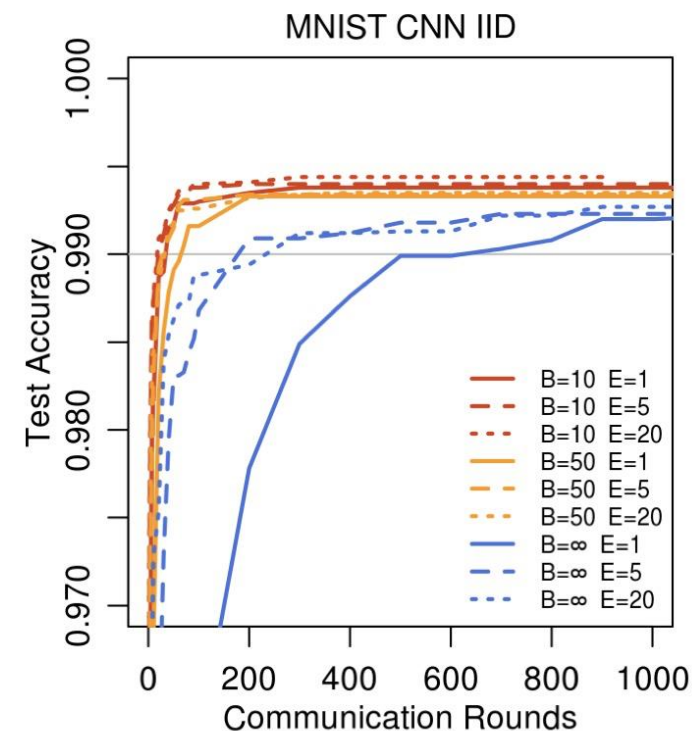
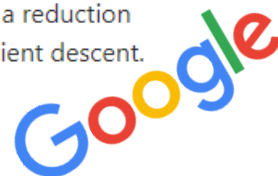
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Target accuracy reached

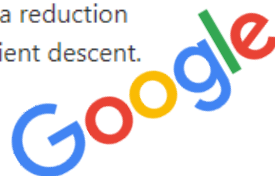
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Next word prediction task

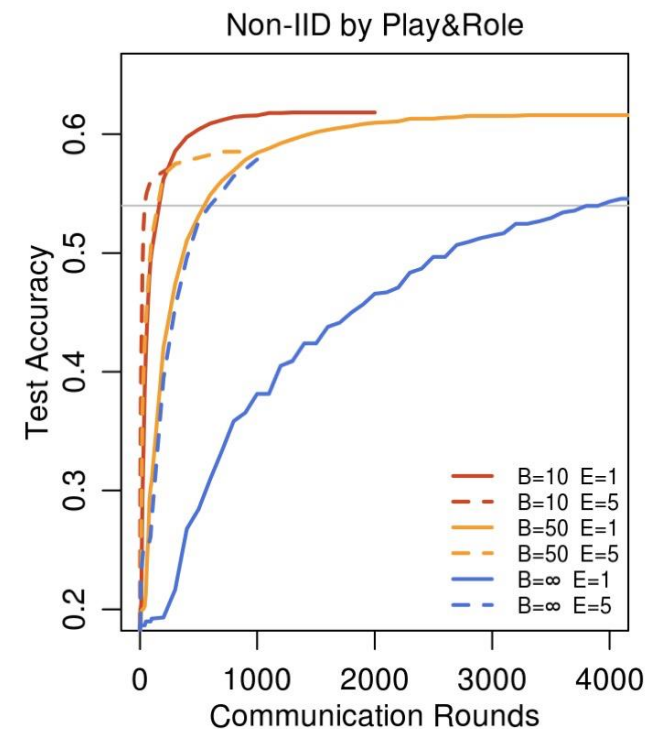
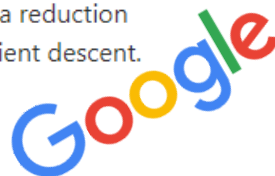
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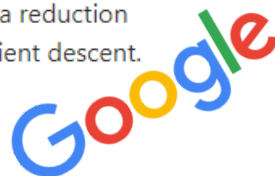
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4 Conclusions and Future Work

Our experiments show that federated learning can be made practical, as FedAvg trains high-quality models using relatively few rounds of communication, as demonstrated by results on a variety of model architectures: a multi-layer perceptron, two different convolutional NNs, a two-layer character LSTM, and a large-scale word-level LSTM.

While federated learning offers many practical privacy benefits, providing stronger guarantees via differential privacy [14, 13, 1], secure multi-party computation [18], or their combination is an interesting direction for future work. Note that both classes of techniques apply most naturally to synchronous algorithms like FedAvg.^[8]

Similar (?) approaches

Distributed Computing

Centralised data, parallel processing, efficiency-focused, identical distribution, no privacy preservation.

Meta-analysis

Decentralised results, statistical aggregation, single-step synthesis, privacy-preserving (no raw data).

Trusted Research Environments (TREs)

Centralised/decentralised storage, secure access, strong governance, restricted data movement, privacy through control not distribution.

Federated Learning

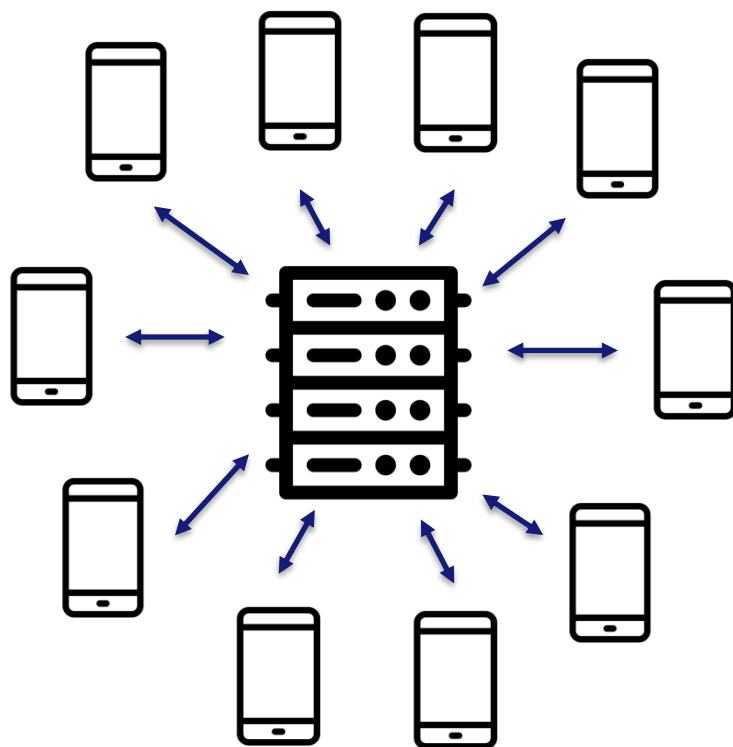
A more detailed characterization

Cross-silo

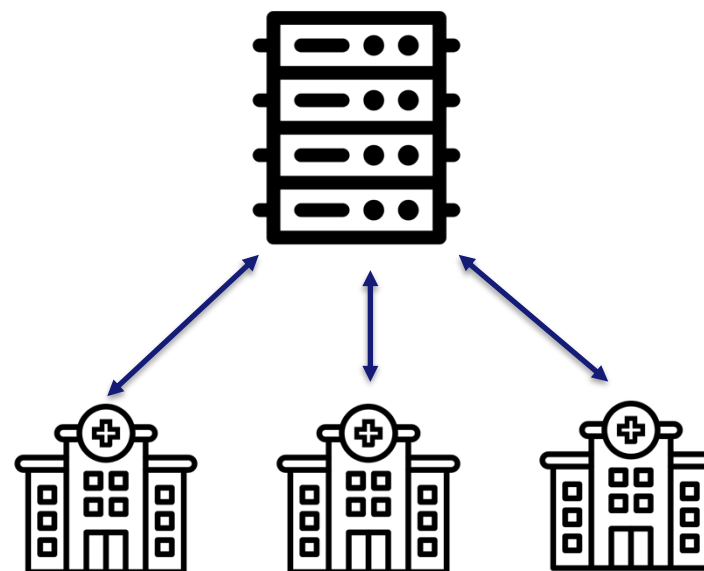
Cross-device

Fields of Use of Federated Learning

Cross-device FL



Cross-silo FL



Fields of Use of Federated Learning

Cross-device FL

EFFICIENCY: sharing models avoids centralizing a potentially extremely large amount of data at a single location.

Example: a model predicting the next word for mobile phones keyboards

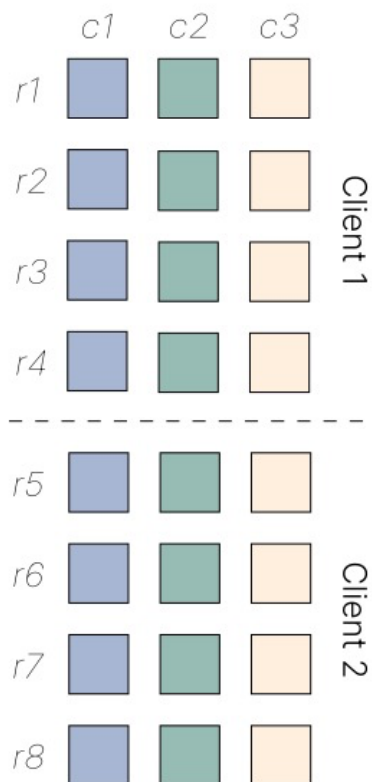
Cross-silo FL

PRIVACY: sharing models presents less privacy issues than sharing data (caveat already mentioned).

Example: a model predicting patients' outcomes

Data partitioning

Horizontally partitioned

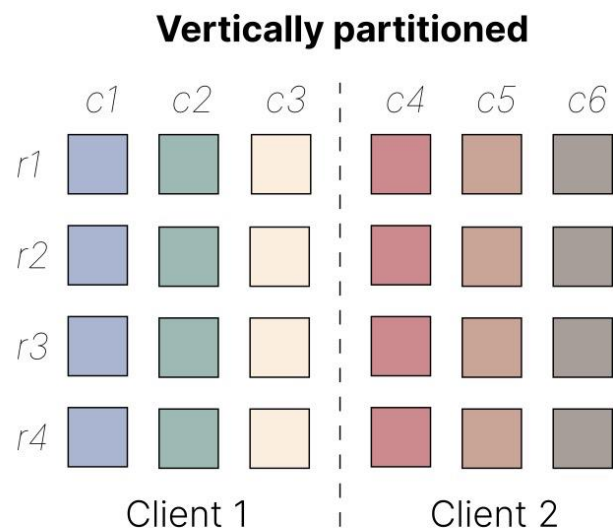


Typically, in bioinformatics, multiple parties hold the **same variables** to describe **different samples**.

For example, multiple institutions may each possess sequencing data from different individuals and conduct a differential gene expression analysis.

Harmonization of data is key!

Data partitioning

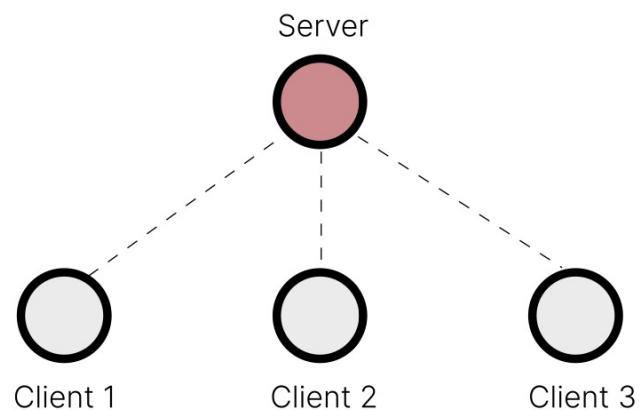


In other domains, multiple parties hold the **same samples** described by **different variables**.

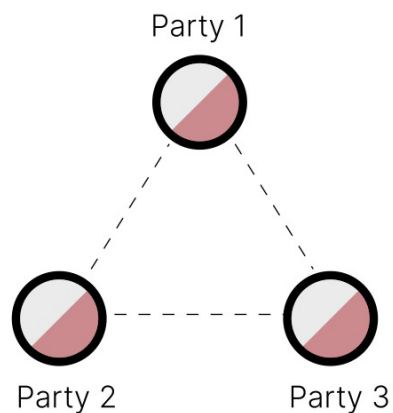
For example, a bank and an insurance company having clients in common may want to collaborate to develop models for targeting their clients.

A central server is not the only way to go

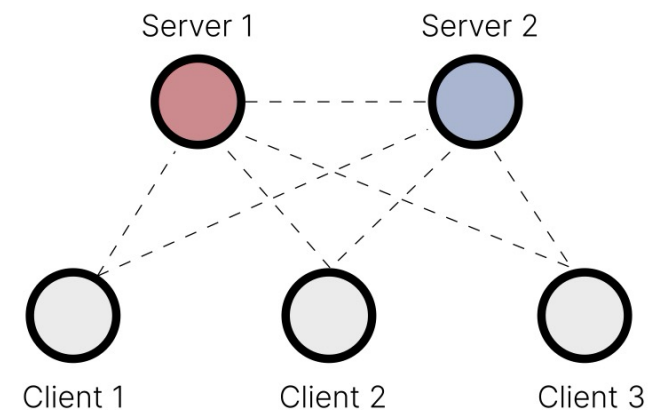
Centralised topology



Decentralised topology

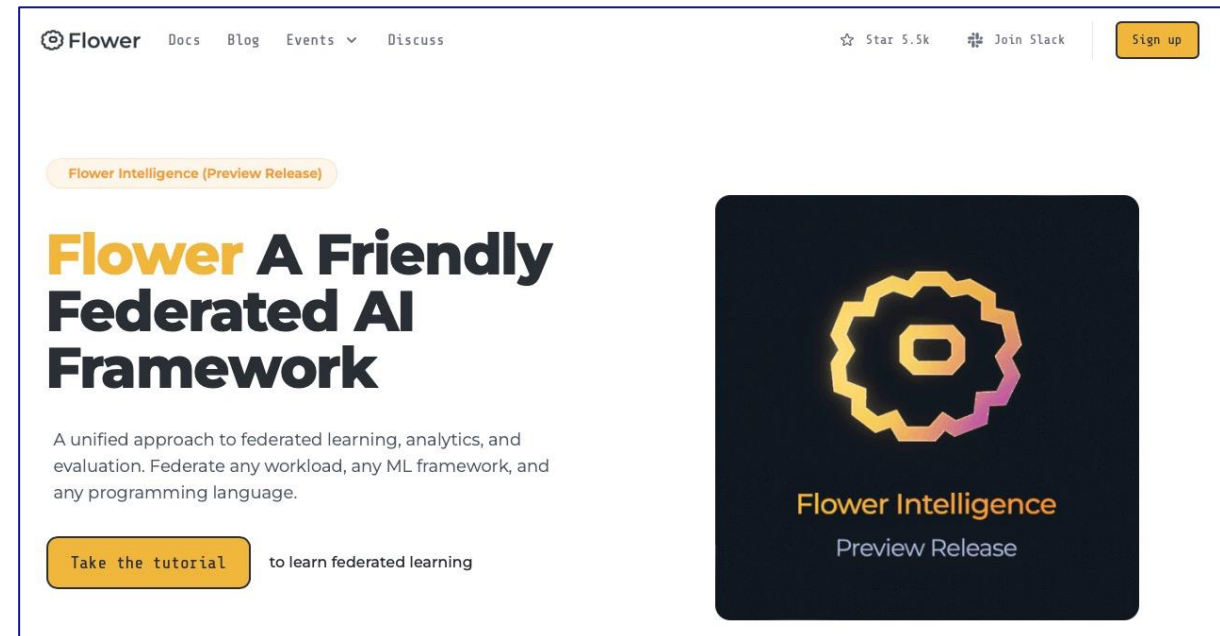


Two-server centralised topology



Software

- Federated learning requires a **framework**: i.e., a software infrastructure that enables the design, deployment, and management of federated learning workflows, handling communication, coordination, security, and aggregation across distributed clients.



Coming next!

Ready to use software for bioinformatics

JOURNAL ARTICLE

sfkit: a web-based toolkit for secure and federated genomic analysis

Simon Mendelsohn , David Froelicher , Denis Loginov , David Bernick , Bonnie Berger  ,
Hyunghoon Cho  [Author Notes](#)

Nucleic Acids Research, Volume 51, Issue W1, 5 July 2023, Pages W535–W541,

<https://doi.org/10.1093/nar/gkad464>

Published: 29 May 2023 **Article history** ▼

<https://sfkit.org>

Published on 12.07.2023 in **Vol 25 (2023)**

 Preprints (earlier versions) of this paper are available at <https://preprints.jmir.org/preprint/42621>, first published September 12, 2022.



The FeatureCloud Platform for Federated Learning in Biomedicine: Unified Approach

Julian Matschinske¹ ; Julian Späth¹ ; Mohammad Bakhtiari¹ ; Niklas Probul¹ ;
Mohammad Mahdi Kazemi Majdabadi¹ ; Reza Nasirigerdeh² ;
Reihaneh Torkzadehmahani² ; Anne Hartebrodt³ ; Balazs-Attila Orban⁴ ;
Sándor-József Fejér⁴ ; Olga Zolotareva² ; Supratim Das¹ ; Linda Baumbach⁵ ;
Josch K Pauling² ; Olivera Tomašević⁶ ; Béla Bihari⁴ ; Marcus Bloice⁷ ;
Nina C Donner⁸ ; Walid Fdhila⁹ ; Tobias Frisch³ ; Anne-Christin Hauschild¹⁰ ;
Dominik Heider¹¹ ; Andreas Holzinger⁷ ; Walter Hötzenndorfer¹² ; Jan Hospes¹² ;
Tim Kacprowski¹³ ; Markus Kastelitz¹² ; Markus List² ; Rudolf Mayer⁹ ;
Mónika Moga⁴ ; Heimo Müller⁷ ; Anastasia Pustozero⁹ ; Richard Röttger³ ;
Christina C Saak¹ ; Anna Saranti⁷ ; Harald H H W Schmidt¹⁴ ; Christof Tschohl¹² ;
Nina K Wenke¹ ; Jan Baumbach¹ 

<https://featurecloud.ai>

Federated Learning

Adversarial attacks and privacy-preserving techniques

How secure is federated learning?

Privacy attacks

- **Property inference:** infer hidden attributes of a client dataset (e.g., disease prevalence)
- **Membership inference:** infer whether a person/sample was in training data
- **Reconstruction / model inversion:** reconstruct sensitive features or records from gradients/updates

Poisoning attacks

- **Data poisoning:** manipulate local training data
- **Model poisoning:** send malicious updates to corrupt the global model
- **Backdoor attacks:** implant hidden malicious behavior triggered by specific inputs

Example of model inversion attack

Reconstructing Training Data from Trained Neural Networks

Niv Haim*

Weizmann Institute of Science
niv.haim@weizmann.ac.il

Gal Vardi*†

TTI-Chicago and Hebrew University
galvardi@ttic.edu

Gilad Yehudai*

Weizmann Institute of Science
gilad.yehudai@weizmann.ac.il

Ohad Shamir

Weizmann Institute of Science
ohad.shamir@weizmann.ac.il

Michal Irani

Weizmann Institute of Science
michal.irani@weizmann.ac.il

(NeurIPS 2022)

Reconstructed images starting from model



Most similar instances in real dataset



Examples from bioinformatics

- **Individual reidentification** (membership inference attack) is an issue for genetic data.
- However, such works often make **unrealistic assumptions** on the level of access to the models and the data (Kolobkov, 2024).

 OPEN ACCESS  PEER-REVIEWED

RESEARCH ARTICLE

Resolving Individuals Contributing Trace Amounts of DNA to Highly Complex Mixtures Using High-Density SNP Genotyping Microarrays

Nils Homer, Szabolcs Szelinger, Margot Redman, David Duggan, Waibhav Tembe, Jill Muehling, John V. Pearson, Dietrich A. Stephan, Stanley F. Nelson, David W. Craig 

Published: August 29, 2008 • <https://doi.org/10.1371/journal.pgen.1000167>

JOURNAL ARTICLE

Deterministic identification of specific individuals from GWAS results

Ruichu Cai , Zhifeng Hao , Marianne Winslett  , Xiaokui Xiao , Yin Yang , Zhenjie Zhang  , Shuigeng Zhou  [Author Notes](#)

Bioinformatics, Volume 31, Issue 11, June 2015, Pages 1701–1707,
<https://doi.org/10.1093/bioinformatics/btv018>

Published: 27 January 2015 **Article history** ▼

Mitigation strategies for adversarial attacks: homomorphic encryption

- Homomorphic encryption lets you perform computations on encrypted data **without** decrypting it.
- The results, once decrypted, are the **same** as if the operations were done on the raw data.
- This enables secure data processing and sharing without exposing sensitive information.

```

from phe import paillier

# Generate public and private keys
public_key, private_key = paillier.generate_paillier_keypair()

# Two plaintext numbers
a, b = 7, 15

# Encrypt both numbers
enc_a = public_key.encrypt(a)
enc_b = public_key.encrypt(b)

# Homomorphic addition: E(a) + E(b) = E(a+b)
enc_sum = enc_a + enc_b

# Direct encryption of the sum
enc_direct = public_key.encrypt(a + b)

# Decrypt both results
print("Decrypt(E(a)+E(b)) =", private_key.decrypt(enc_sum))
print("Decrypt(E(a+b))      =", private_key.decrypt(enc_direct))

```

[1] ✓ 0.2s

```

... Decrypt(E(a)+E(b)) = 22
      Decrypt(E(a+b))  = 22

```

Preservation of addition in Paillier scheme.

Mitigation strategies for adversarial attacks: homomorphic encryption

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✓ Advantages

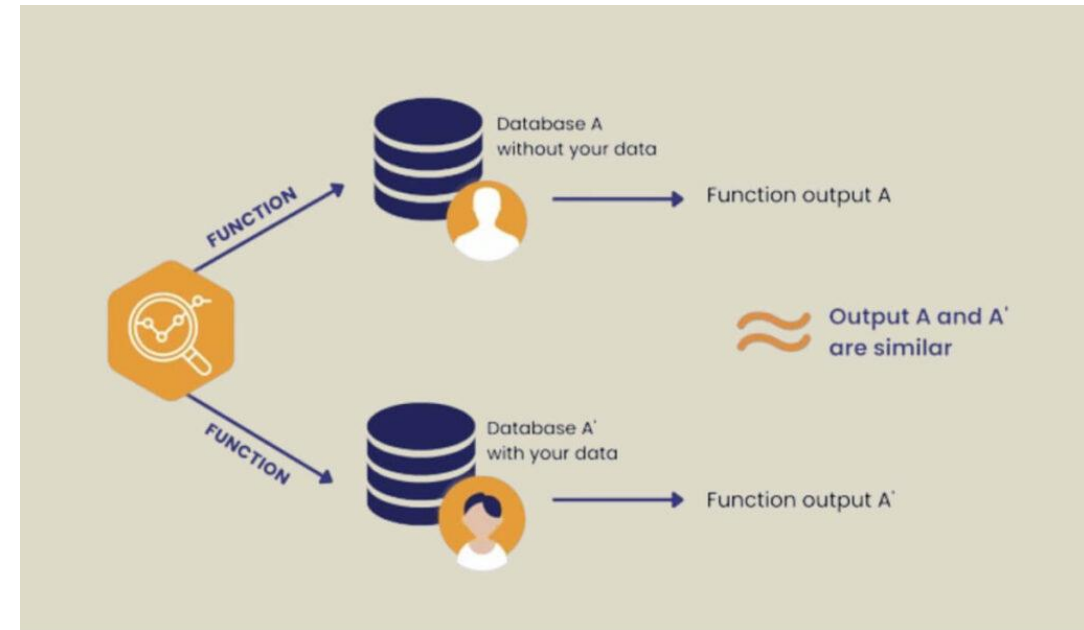
- Data stays encrypted during processing
- Provides strong privacy protection

✗ Disadvantages

- Much slower than normal computation (high computational cost)
- Requires more storage and bandwidth due to large ciphertexts

Mitigation strategies for adversarial attacks: differential privacy

- Differential privacy protects individuals by adding carefully **calibrated** random **noise** to data or query results.
- This ensures that no **single person's** information can be distinguished with high confidence.
- It balances data utility with strong **mathematical privacy guarantees**.



<https://pvml.com/glossary/differential-privacy/>

Mitigation strategies for adversarial attacks: differential privacy

- Differential privacy protects individuals by adding carefully **calibrated** random **noise** to data or query results.
 - This ensures that no **single person's** information can be distinguished with high confidence.
 - It balances data utility with strong **mathematical privacy guarantees**.
- ✓ **Advantages**
 - Provides formal, provable privacy guarantees
 - Scales well for large datasets
 - ✗ **Disadvantages**
 - Noise can reduce accuracy of results
 - Choosing the privacy budget (*epsilon*) is difficult
 - Less effective for small datasets

Mitigation strategies for adversarial attacks: secure multiparty computation (SMPC)

- SMPC allows multiple parties to **jointly** compute a result without revealing their private inputs.
- Each participant only learns the **final output**, not others' data.

Example: privacy-preserving sum in FL

In this simple example, three clients, with values 5, 10, and 15, respectively, aim to securely calculate their sum, which has a true value of $5 + 10 + 15 = 30$. We show how to compute this sum using three techniques described in Section 2.6.

Secure Multiparty Computation

- Clients split their values into random shares as $\{2; 1; 2\}$, $\{3; 4; 3\}$, and $\{5; 5; 5\}$ respectively, and then send the first two shares each to one of the other two clients.
- Clients sum the received shares and their local share to obtain 10, 9, and 11 respectively, and then send the obtained values to the server.
- The server sums the received values, obtaining 30.

Mitigation strategies for adversarial attacks: secure multiparty computation (SMPC)

- SMPC allows multiple parties to **jointly** compute a result without revealing their private inputs.
- Each participant only learns the **final output**, not others' data.

Advantages

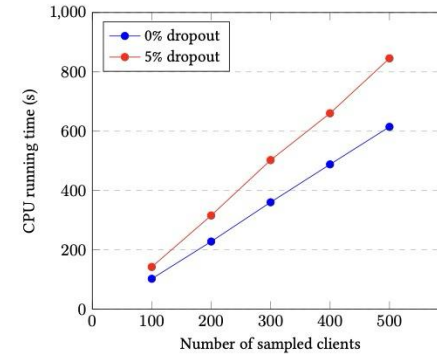
- Strong privacy: inputs remain secret throughout the computation
- Cryptographic guarantees of correctness and confidentiality

Disadvantages

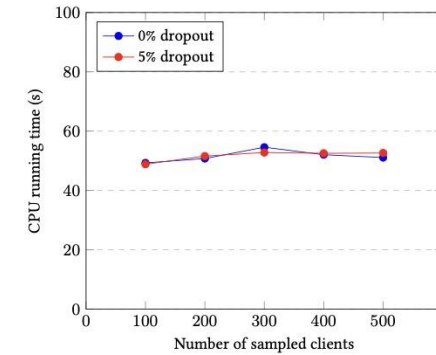
- Computationally and communication intensive (slower than normal computing)
- Protocols can be complex to design and implement securely
- Scalability challenges

SecAgg+ (provided by Flower as a workflow!)

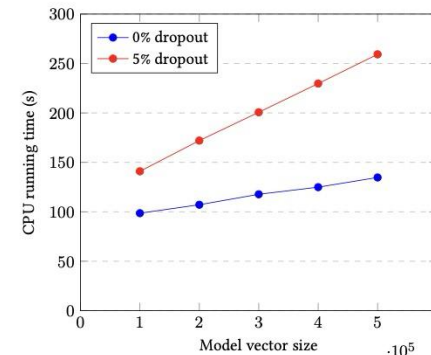
- **Efficient graph-based mask exchange:** clients only need to communicate with a small subset of peers.
- **Robustness to dropouts:** the protocol is designed to handle client dropouts without redoing heavy computations.



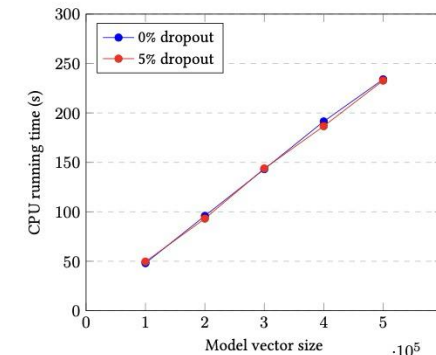
(a) Running time of server with increasing number of clients



(b) Running time of client with increasing number of clients



(d) Running time of server with increasing vector size



(e) Running time of client with increasing vector size

Federated Learning

Legal considerations

Key points on legal considerations

Data protection still required: FL is legally treated like regular data sharing under GDPR, there are no specific regulations (yet).

Consortium agreements: Parties must sign Data Protection Agreements defining roles, responsibilities, interactions, ethics approvals, and handling of privacy breaches.

Intellectual property & third parties: Joint control requires agreements on IP rights, patents, and possible financial gains.

Patient rights: In several countries, patients need to be able to withdraw consent at any time, meaning technical/legal procedures must exist to remove their data from future analyses.

FL and GDPR principles

- **Data minimisation (Art. 5(1)(c))**
Share model updates/parameters instead of transferring full datasets.
- **Storage limitation (Art. 5(1)(e))**
Less need to create large central copies of personal data.
- **Integrity and confidentiality (Art. 5(1)(f))**
Reduced raw-data movement can lower exposure risk (especially privacy-enhancing methods).
- **Privacy by design and by default (Art. 25)**
FL, under certain conditions, can be an architecture that embeds privacy protections from the start.

Final considerations

- **Technology evolves faster than regulation:** dedicated legal frameworks for federated learning are likely to emerge in the coming years.
- **Objective perspective:** federated learning is often more aligned with key GDPR principles than centralized data-sharing.
- **Subjective perspective:** federated learning can make institutions, partners, and patients more comfortable participating in collaborative studies. By reducing direct data sharing and strengthening protections against data leakage, federated learning can lower the perceived risk of joining a consortium.

Federated Learning

Resources

The main source for this presentation



Technical and legal aspects of federated learning in bioinformatics: applications, challenges and opportunities

Daniele Malpetti^{1,2†}, Marco Scutari^{1*†}, Francesco Gualdi^{1,2†},
Jessica van Setten³, Sander van der Laan^{4,5}, Saskia Haitjema⁴,
Aaron Mark Lee⁶, Isabelle Hering⁷ and Francesca Mangili^{1,2}

¹Istituto Dalle Molle di Studi sull'Intelligenza Artificiale (IDSIA), USI-SUPSI, Polo Universitario Lugano, Lugano, Switzerland, ²Swiss Institute of Bioinformatics (SIB), Lugano, Switzerland, ³Department of Cardiology, University Medical Center Utrecht, University of Utrecht, Utrecht, Netherlands, ⁴Central Diagnostics Laboratory, University Medical Center Utrecht, University of Utrecht, Utrecht, Netherlands, ⁵Department of Genome Sciences, University of Virginia, Charlottesville, VA, United States, ⁶William Harvey Research Institute, NIHR Barts Biomedical Research Centre, Queen Mary University of London, London, United Kingdom, ⁷Étude Hering, DPO Associates SARL, Nyon, Switzerland

7 Conclusions

Independent research efforts in several bioinformatics domains have shown federated learning to be an effective tool to improve clinical discovery while minimising data sharing (10, 12, 13, among others). FL enables access to larger and more diverse data pools, resulting in faster and more robust exploration and interpretation of results. At the same time, it provides enhanced data privacy and can seamlessly incorporate advanced, encrypted and secure computation techniques. Despite the increased computational requirements and reduced ability to explore and troubleshoot issues with the data (104), the benefits of FL may outweigh these additional costs.

Therefore, federated learning can potentially mitigate the risks associated with national regulations if implemented in a manner that is secure by design and by default. Its use may also make patients and institutions more confident in participating in clinical studies by reducing privacy and data misuse risks. However, the reliable use of federated learning and its effective translation into clinical practice require a concerted effort by machine learning, clinical, and information technology specialists. All their skills are necessary to accurately evaluate the associated risks and expand its practical applications in bioinformatics beyond the early-stage applications reviewed in this paper.

Questions?

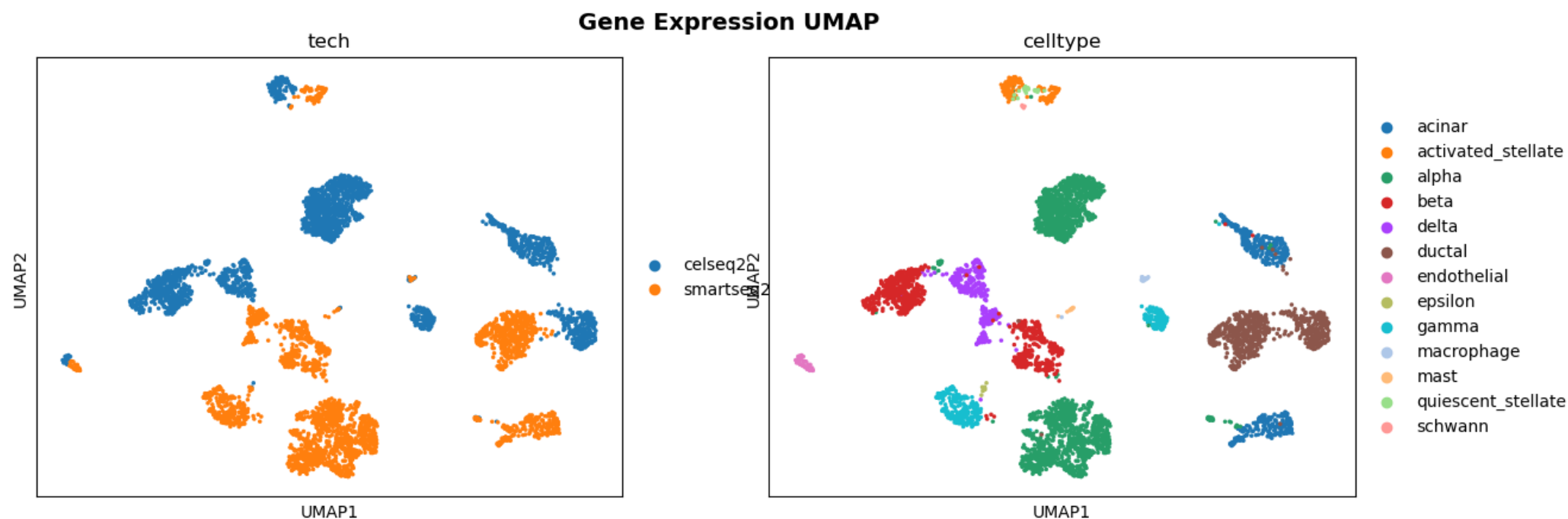
Today's practical

Overview

These practicals emulate, in simplified form, a realistic collaborative workflow. Seven institutions worldwide each hold pancreas single-cell datasets, with each dataset generated using a different sequencing technology. They collaborate to train an open-source model for technology-related batch-effect removal, using an architecture provided by **scVI-tools**, and plan to disseminate it through a journal publication. Since data cannot be shared across sites, the partners adopt **Federated Learning (FL)** with the **Flower** framework, allowing training to occur locally while only model updates are exchanged.

A separate institution, which also has pancreas single-cell data, later reads the publication and wishes to use the released model. This is particularly useful for them because their study contains measurements from two different technologies (one machine failed mid-study and was replaced), and technology-related batch effects are present. They therefore need to remove these batch effects to perform several downstream tasks.

Our starting point: batch effects need to be removed before downstream tasks



Task 1: identifying highly variable genes

